EXECUTION FLOW CONCEPT MAP

Shows how a USER Program request is executed using System Calls from User mode to Kernel

USER PROGRAM

user writes and runs a program in user mode

Application programming interface

Provides functions (library calls) to request OS services

System call interface

Interface (eg. trap instruction switches control from user mode to kernel mode

OPERATING SYSTEMS

Kernel

Kernel validates the request performs required operations using OS services

Hardware

CPU, Memory, Devices execute the operation or provide data.

Response Back to user

Result is returned from kernel to user program through system calls interface and API

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System Calls act as a bridge allowing user programs to request OS services

SYSTEM CALLS

Interface between User Programs and Operating System Services

PROCESS SYSTEM CALLS

Manage process creation, execution, termination & control

Examples

- fork()
- exec()
- exit()
- wait()

FILE SYSTEM CALLS

Manage files and directories on storage devices

Examples

- open()
- read()
- close()
- unlink()

DEVICE SYSTEM CALLS

Request I/O operations & control device behaviour

Examples

- read()/write()
- ioctl()
- open()/close()
- get/set

INFORMATION SYSTEM CALLS

Get or set system information & attributes

Examples

- getpid()
- time()
- chdir()
- uname()

RELATION TO OS SERVICES

SCHEDULING

Decides which process runs next

Used by: Process system calls.

MEMORY ALLOCATION

Allocates and manages memory for processes

Used by: All system call types

DEVICE HANDLING

Controls I/O devices via drivers and controllers

Used by: Device System calls

Protection & Security

Ensures safe access to resources & permission

Used by: All system call types